



Installation Instruction

Installation Instruction Number: 5414639

Released Date: 04-dec-2017

Aftertreatment Outlet NOx Sensor High Temperature Sealant Installation Instructions

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Introduction

The purpose of this document is to outline how to install high temperature sealant around the NOx sensor drip shield in the aftertreatment selective catalyst reduction (SCR) catalyst.

Product Affected

- X15 CM2350 X114B
- X15 CM2350 X116B

Service Parts

Required part numbers are shown in Table 1 below.

Table 1, Service Parts		
Description	Part Number	Quantity
High temperature sealant	5362108	1

Cummins® High temperature sealant, Part Number 5362108, is the **only** approved sealant for this procedure.

Personal Protection Equipment (PPE)

- Completing a Job Safety Assessment (JSA) prior to performing work helps identify job safety hazards and prevent incidents.
- To reduce possibility of personal injury, personal protective equipment (PPE) must be utilized. Reference General Safety Instructions Procedure 204-006 in Section I of corresponding Service Manual

Required Tools

- Sand paper 40 to 60 Grit
- Masking tape
- Safety glasses
- Rubber gloves
- Stir stick
- Flat tip screwdriver
- Needle nose pliers
- Chemical free cleaning solvent
- Cleaning brush

- Shop towels

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect battery. See equipment manufacturer service information.



Sensor Removal

⚠ WARNING ⚠

Wait for aftertreatment system to cool down before working on hot exhaust or exhaust components. Failure to do so could result in personal injury.

- Remove original equipment manufacturer (OEM) exhaust stack and expose aftertreatment SCR outlet.
- Remove:
 - NOx sensor: See corresponding Service Manual. Reference Procedure 019-451 in Section 19
 - Particulate matter sensor: See corresponding Service Manual. Reference Procedure 019-478 in Section 19
 - Exhaust gas temperature sensor: See corresponding Service Manual. Reference Procedure 019-449 in Section 19
- Store all sensors in a safe place.



Surface Preparation and Masking

⚠ CAUTION ⚠

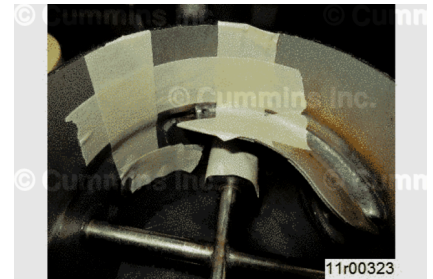
Do not use any chemical solvents or cleaners to clean internal metal surface. Possible damage to aftertreatment SCR catalyst can occur.

- Score circled surfaces shown in figure with 60 grit sand paper.



- Verify all aftertreatment SCR catalyst surfaces are clean by using a chemical free solvent with a clean shop cloth.

- Apply masking tape on portions as shown in figure.
- Particulate matter sensor port, exhaust gas temperature sensor, and meshed area of NOx sensor cup **must** be completely covered.



High Temperature Sealant Application

- All surfaces **must** be free of oil, grease, dirt, corrosion, or other contaminants before using high temperature sealant.
- High temperature sealant **must** be mixed thoroughly for at least one minute.
- While wearing rubber gloves, use your finger or an application tool to apply high temperature sealant as shown in figure directly above.



- Minimize amount of high temperature sealant that falls into aftertreatment SCR catalyst during application. Figure shows desired application of high temperature sealant. A single 1 ounce container may be enough for five or more individual applications.
- Apply sealant to masked area and verify press fit wing and top of rain shield is covered as shown in figure.
- Verify high temperature sealant is well packed, all air pockets are removed, and bonding is concentrated to masked area.
- **Not** shown in figure: high temperature sealant will need to be applied to top of NOx sensor cup, under water shield.
- Use caution when applying under shield as sealant may fall into SCR Outlet.
- Application of sealant will **not** be directly visible to repairer. Use index finger and trace underside of shield.



- High temperature sealant **must** applied to top portion of NOx cup to alleviate any path for water to intrude.
- Starting from center of water shield, apply sealant along seam of NOx sensor cup and water shield. Verify sealant is packed and smoothed across entire seam of shield as shown in figure.

- Figure illustrates finished product and an approximate amount of sealant used to complete repair.
- Remove all masking tape and verify section on NOx sensor screen is last. This will reduce any excess sealant falling into NOx sensor wheel.



- Remove tape before drying begins.
- Sealant cure time is two hours depending on application thickness.
- See figure for example of completed high temperature sealant installation.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- The high temperature sealant cures at room temperature and is fully cured four hours after application to surface. according to manufacturer's instructions.
- Install:
 - NOx sensor: See corresponding Service Manual. Reference Procedure 019-451 in Section 19
 - Particulate matter sensor: See corresponding Service Manual. Reference Procedure 019-478 in Section 19
 - Exhaust gas temperature sensor: See corresponding Service Manual. Reference Procedure 019-449 in Section 19
- Connect batteries. See equipment manufacturer service information.
- After high temperature sealant has cured, install OEM exhaust stack.
- Perform a stationary regeneration. See corresponding Service Manual. Reference Procedure 014-016 in Section 14.

Document History

Date	Details
2017-12-4	Module Created

Last Modified: 08-Dec-2017
